KAVIYA MANICKAVASAGAM PANNEERSELVAM

kaaviyamp@gmail.com | 0481 791 055
https://www.linkedin.com/in/kaviya-manickavasagam-panneerselvam-21aa711b2/

CAREER OBJECTIVE

Cybersecurity graduate with hands-on experience in security operations, vulnerability assessment, incident response, and access control reviews, supported by knowledge of industry frameworks including ISO/IEC 27001:2022, NIST CSF, SOCI and ACSC Essential Eight. Skilled in log analysis, threat detection, secure SDLC practices, and governance, risk, and compliance (GRC) principles. Completed applied research projects in IoT security, machine learning–based anomaly detection, and smart contract vulnerability auditing. Seeking an entry-level cybersecurity role to contribute to improving organisational security posture and growth within security domains.

EXPERIENCE

Cyber Security Intern, Le Souk, Adelaide **Dec 2024 - May 2025**

- Designed and distributed a short survey to understand staff and business awareness of secure online practices related to restaurant systems (e.g., booking platforms and online ordering portals), aligned with ISO 27001:2022 information security control principles.
- Analysed responses to identify knowledge gaps in phishing awareness, unsafe link handling, password hygiene, and protection of customer information.
- Reviewed access permissions, data handling practices, and privacy settings within reservation platform, recommending secure login practices and improved protection of customer information.
- Recommended short-term and long-term security controls and awareness measures to reduce cyber risks and support the Confidentiality, Integrity, and Availability (CIA) of business and customer data.

Cyber Security Analyst Trainee, Cognizant, India **Sep 2023 - Dec 2023**

- Analysed security logs and events to detect potential threats using company-standard playbooks and established incident response procedures.
- Assisted in vulnerability assessments and basic threat analysis activities. Supported in the development of an incident response plan based on NIST guidelines (SP 800-61) and preparation of security reports.
- Contributed to internal dashboards using Power BI and reporting mechanisms to monitor operational and security risks.

Network Security Intern, Bharat Sanchar Nigam Limited, India **Feb 2022 - Mar 2022**

- Applied network security fundamentals including TCP/IP, routing and switching, and secure network architecture principles.
- Analysed network traffic and configurations to identify vulnerabilities, misconfigurations, and potential security risks.
- Gained hands-on exposure to firewall concepts, access control mechanisms, network segmentation, and monitoring to support availability and integrity of network services.
- Developed understanding of common network threats such as unauthorized access, interception, and denial-of-service attacks, and supported documentation of security controls aligned with CIA principles.

EDUCATION

- **Master of Cyber Security:** The University of Adelaide **Feb 2024 - Dec 2025**
  Recipient of the Global Citizens Scholarship
- **Bachelor of Electronics & Communication Engineering:** Saranathan College of Engineering **Jun 2019– Mar 2023**
  Published research: “Pipelined Half-Precision Floating Point ALU” in IJARSCT, Volume 3, May 2023

TECHNICAL SKILLS

Programming and OS: C Python, Basic Java, C#, ASP.NET MVC, CSS, SQL, Linux and Windows

Cyber Skills: Identity and Access Management, Secure SDLC, IP Networking & Routing Protocols, Packet Forwarding Architecture, IoT Security, Log Analysis & Event Monitoring, Incident Response, Threat Analysis & Attack Vector Identification, Vulnerability Assessment & Mitigation, Security Policies & Control Implementation.

Cybersecurity Frameworks: ACSC Essential 8, NIST Cybersecurity Framework (CSF) 2.0, ISO 27001:2022

Tools & Methodologies: Splunk, Power BI, MS Excel, Agile, SDLC

AI & ML: Supervised and unsupervised learning, anomaly detection, security analytics

CYBERSECURITY PROJECTS AND RESEARCH

Smart Contract Vulnerability Detector using LLM (Research) **Aug 2025 - Dec 2025**

- Built a hybrid smart contract auditing system combining regex-based static analysis with Google Gemini LLM to detect critical Solidity vulnerabilities.
- Developed a five-stage JSON repair and validation pipeline to ensure consistent outputs and reliable audit reporting Applied TF-IDF and Random Forest multi-label classification to automate vulnerability detection and improve accuracy.
- Created a Streamlit-based application for real-time contract analysis, visualization, and downloadable audit reports.
- Evaluated SmartBugs-Wild contracts to optimize detection performance, reduce false positives, and enforce secure coding standards.

Man-in-the-Middle Attacks on MQTT-based IoT using Adversarial Machine Learning **Sep 2025 – Dec 2025**

- Designed and implemented an adversarial testing framework for MQTT-based IoT systems to simulate Man-in-the-Middle (MITM) attacks and evaluate communication security.
- Applied machine learning models to detect anomalous MQTT traffic, assessing the impact of adversarial perturbations on intrusion detection performance.
- Conducted controlled experiments to analyse attack effectiveness, measuring false positives and false negatives under adversarial conditions.
- Documented findings and insights on improving IoT security resilience using context-aware and ML-based detection techniques.

Safe Home Service **Aug 2025 – Dec 2025**

- Designed and developed a secure full-stack web application (MERN stack) with a strong focus on privacy-by-design and secure handling of user data. Integrated security controls for authentication and authorisation.
- Implemented decentralised, privacy-preserving contact tracing mechanisms, reducing centralised data storage and limiting personal data exposure.

CERTIFICATIONS

- (ISC)²: Certified in Cybersecurity (CC)
- Delloite and Datacom Stem Stimulation
- Google Certified: Fundamentals of Cyber Security
- Google Certified: Connect and Protect: Networks and Network Security
- Google Certified**:** Play It Safe: Manage Security Risks
- Python, Web Development, Agile
- Emotional Intelligence for Leadership
- Power BI

REFERENCE

Steve Balaji
Portfolio Analyst- PMO
DHW - Digital Health SA
Steve.Balaji@sa.gov.au